

# Grounded RAG Pipeline for Code-Switching (Hinglish) in Clinical Decision Support Systems.



---

GROUP 5 | PROF. DR. YOGESH NAIK | PROF. DR. RAJESH MAURYA

**01**

# **Problem Statement & Relevance.**

---

# the problem statement...

Modern clinical AI systems assume that medical reasoning **begins with clean, formal language**.

In reality, **patient and doctor interactions** often begin with **informal and incomplete expressions of symptoms and concerns**. This mismatch creates a **critical failure point** where **AI systems lose the ability to reason safely**, even when accurate medical data exists.

In **real-world clinical settings**, especially in **India**, patient and doctor interactions often begin with such **informal code-switched expressions**, commonly referred to as **Hinglish**, where Hindi and English are **interwoven**, such that it :

- lacks standardized **grammar**
- **varies** by region, education
- context **encodes** medical meaning **indirectly**

Report: *"Right-sided pleural effusion"*

Query: *"Doctor, right side chhati me pani bhar gaya hai kya?"*

**Thus, this is not a translation problem. It is a reasoning alignment problem.**

## *the problem statement...*



Clinical information, however, is recorded in **formal English medical terminology**, creating a **semantic disconnect** at the point of interpretation when patients **communicate in Hinglish**. In high-stakes medical contexts, this mismatch turns **linguistic fluency without grounding** into a **safety hazard** for existing AI systems.

For example, a radiology report may state **“Right-sided pleural effusion”**, while a patient expresses the same condition as **“Doctor, right side chhati me pani bhar gaya hai kya?”** Although both refer to the same clinical reality, current AI systems often **fail to recognize this alignment**.

Such Hinglish expressions are not simple translation variants of English medical terms but they **encode clinical intent** in **culturally grounded and indirect ways**. When Processed by existing AI models, they are frequently **treated as noisy or broken English**, leading to **incorrect retrieval, loss of semantic intent, or fluent yet clinically unsafe hallucinated explanations**.

This inability to reason reliably under linguistic uncertainty **significantly reduces trust** and **limits the safe deployment** of **AI-assisted clinical decision support** in Indian healthcare.

**02**

## **Proposed Scope.**

---

## *the proposed scope...*

This research proposes the design and evaluation of a **grounded Retrieval-Augmented Generation (RAG) pipeline** for clinical decision support under **code-switched Hinglish queries**.

The system aligns **informal Hinglish clinical intent** with **verified English clinical case narratives** from a multi-specialty open-access corpus, as a **reliable source of medical information**. LaBSE cross-lingual encoding, FAISS retrieval, and adaptive passage truncation **yield ranked context that strengthens grounding** in generation.

Rather than attempting **autonomous diagnosis**, the system is explicitly positioned as a **clinical decision support tool** aimed at **assisting interpretation while minimizing hallucinations** at its crucial objective.

## *the proposed scope...*

The scope is intentionally controlled to focus on **Hinglish as the linguistic setting**, explanation generation as the primary output, and **evidence-grounded reasoning** as the central research objective.

**Retrieval-based grounding**, through cross-lingual text encoding, dense similarity retrieval, and adaptive selection of retrieved passages, aims to keep **generated Hinglish explanations factually consistent** with ranked English clinical case narratives, supporting safer and more **reliable use in healthcare settings**.

Within this scope, the research aims to **contribute Hinglish–English query–evidence pairs**, a cross-lingual retrieval approach linking informal phrasing to case-level medical text, and **an evaluation focus on clinical factuality** and evidence alignment, **not surface fluency alone**.

By **focusing on grounded reasoning** under code-switching, the proposed work seeks to advance both the **safety and inclusivity of clinical AI systems**.

**03**

# **Literature Review.**

---



# *the literature review...*

- 1) Thakur, R., Sharma, S., & Chopra, G. (2025). HiFACTMix: A Code-Mixed Benchmark and Graph-Aware Model for EvidenceBased Political Claim Verification in Hinglish. arXiv preprint arXiv:2508.10001.
- 2) Ghosh, A., Acharya, A., Saha, S., Pandey, G., Raghu, D., & Sinha, S. (2024, November). Healthalignsumm: Utilizing alignment for multimodal summarization of code-mixed healthcare dialogues. In Findings of the Association for Computational Linguistics: EMNLP 2024 (pp. 11546-11560).
- 3) Ghosh, A., Acharya, A., Jha, P., Gaudgaul, A., Majumdar, R., Saha, S., ... & Agarwal, S. (2024). MedSumm: A Multimodal Approach to Summarizing Code-Mixed Hindi-English Clinical Queries. arXiv preprint arXiv:2401.01596.
- 4) Xia, P., Zhu, K., Li, H., Wang, T., Shi, W., Wang, S., ... & Yao, H. (2024). Mmed-rag: Versatile multimodal rag system for medical vision language models. arXiv preprint arXiv:2410.13085.
- 5) Gumma, V., Raghunath, A., Jain, M., & Sitaram, S. (2024). Health-pariksha: Assessing rag models for health chatbots in real-world multilingual settings. arXiv preprint arXiv:2410.13671.
- 6) Ke, Y. H., Jin, L., Elangovan, K., Abdullah, H. R., Liu, N., Sia, A. T. H., ... & Ting, D. S. W. (2025). Retrieval augmented generation for 10 large language models and its generalizability in assessing medical fitness. npj Digital Medicine, 8(1), 187.



**04**

## **Research Gap.**

---

# the research gap...

**paper 1]** HiFACTMix introduces HiFACT, a benchmark of approximately 1,500 real-world Hinglish claims annotated with textual evidence and veracity labels, and proposes a retrieval-augmented, graph-aware framework for evidence-linked explanation generation under code-mixing. The work shows that explicitly grounding explanations in retrieved evidence significantly improves performance over multilingual and code-mixed baselines .

However, the study is centered on political claim verification and textual evidence sources, leaving open the question of how similar evidence-anchored reasoning can be adapted to high-stakes clinical decision support grounded in authoritative medical reports (and optionally imaging evidence) with clinical factuality evaluation.

---

**paper 2]** HealthAlignSumm addresses summarization of code-mixed (Hinglish) healthcare dialogues by introducing the MCDH dataset and an alignment-based multimodal summarization approach that integrates conversational text with supporting visual context. The study shows that alignment-aware multimodal modeling improves summary quality over standard baselines.

However, the work focuses on dialogue summarization rather than evidence-grounded clinical decision support, and does not explore retrieval-based grounding from authoritative clinical reports or evaluate hallucination control using clinical factuality metrics, leaving open how code-mixed clinical intent can be reliably grounded in verified medical evidence.



# the research gap...

**paper 3]** MedSumm: A Multimodal Approach to Summarizing Code-Mixed Hindi-English Clinical Queries

This paper highlights that **existing research** in medical question summarization **has primarily concentrated on text-based methods, overlooking the critical integration of visual cues** (such as symptom photos) that patients often provide.

Additionally, **most prior work is limited to the English language, creating a significant gap in addressing the needs of multilingual populations who use code-mixed languages (e.g., Hindi-English)** in low-resource settings. Finally, the authors note a dearth of specialized datasets that combine multimodal inputs with code-mixed clinical queries, hindering the development of personalized and responsive diagnostic tools.

---

**paper 4]** MMed-RAG: Versatile Multimodal RAG System for Medical Vision Language Models

The **authors identify** that current Retrieval-Augmented Generation (**RAG**) **implementations for medical models are often dataset-specific**, limiting their generalizability across diverse medical domains like radiology, pathology, and ophthalmology.

**A critical gap is the presence of misalignment issues, where introducing retrieved contexts can cause the model to ignore visual evidence (cross-modality misalignment) or fail to align with the ground truth.**

Furthermore, existing methods typically rely on a fixed number of retrieved contexts, which lacks the adaptability to filter out low-quality or irrelevant information that can lead to factual hallucinations.



# the research gap...

**paper 5]** Although **HEALTH-PARIKSHA** provides a **structured evaluation of RAG models** for multilingual health chatbots, it does not fully **capture real-world linguistic and cultural diversity**. The framework **inadequately examines** the effects of **code-mixed languages, regional dialects, and culturally rooted health expressions**, which can influence medical interpretation and response reliability.

Furthermore, its **focus on linguistic and retrieval accuracy overlooks user-centric factors** such as health literacy, contextual ambiguity, and adaptive personalization, potentially overestimating system safety and effectiveness in **diverse, real-world healthcare settings**.

---

**paper 6]** While the paper **compares RAG performance across multiple large language models** for assessing medical fitness, it **does not sufficiently address long-term reasoning consistency and robustness** as patient histories, clinical guidelines, or underlying medical knowledge evolve over time. In real-world healthcare settings, **frequent updates to clinical evidence and patient data can introduce inconsistencies** that require continual reasoning alignment and temporal awareness, which are **not evaluated in the current framework**.

Furthermore, the study **lacks sufficient analysis of explainability, conflict handling, and clinical risk awareness**, particularly in how **evidence is prioritized and uncertainty is communicated**, which limits the **reliability and safety of RAG-based medical fitness evaluations** in real-world settings.



**05**

# **Hypotheses / Conjectures.**

---



# hypotheses formulation...



Based on the literature review and observed limitations of existing clinical AI systems under code-switched inputs, we hypothesize that **evidence-grounded generation is essential for safe and reliable clinical decision support** in Hinglish settings.

In particular, **retrieval-augmented generation (RAG)**, anchored in authoritative medical text evidence, is expected to **mitigate hallucinations** and **preserve factual consistency** even when linguistic input is informal or code-switched.

## Statistical Hypotheses:

**H<sub>01</sub> (Grounding Effect):** there is no significant difference in factual consistency or hallucination rate between Hinglish explanations generated by zero-shot LLMs & evidence-grounded RAG systems.

**H<sub>02</sub> (Code-Mixing Robustness):** increasing the level of Hindi–English code-mixing does not significantly affect the performance difference between grounded and non-grounded models.

**H<sub>03</sub> (Evidence Type):** using authoritative medical text evidence does not significantly improve the factual correctness of Hinglish explanations compared to general biomedical text.

## testing of the hypotheses...

Using an initial prototype built, **H1 and H2 were evaluated with paired comparisons** on the same Hinglish queries: for each query, outputs from **(i) zero-shot LLM** and **(ii) text-grounded RAG**.

**Per-sample scores** were computed for **factual consistency** (evidence-supported clinical concepts) and **hallucination rate** (concepts in the output not supported by retrieved evidence).

H3 (authoritative vs general evidence sources) is deferred to the next phase of the project.

### Statistical Testing Plan:

- **For H1 (RAG vs zero-shot):** mean paired differences were tested with the Wilcoxon signed-rank test; Cohen's d and a 95% confidence interval on the factual gain were reported.

- **For H2 (code-mixing):** samples were grouped by Code-Mixing Index (low / medium / high tertiles). Kruskal–Wallis assessed differences across groups; Mann–Whitney U tests with Bonferroni correction were used for pairwise comparisons; Spearman correlation related CMI to grounding outcomes.

For H1 and H2, **we reported p-values, effect sizes, and confidence intervals** (where applicable) to **capture both statistical significance and practical impact**. H3 will follow the same reporting convention when implemented.





## results of the hypotheses...

Table 1. H1: Grounded RAG vs zero-shot (n = 1,165 paired queries)

| Metric              | Zero-shot mean | Grounded mean | Wilcoxon $p$ | Cohen's $d$ |
|---------------------|----------------|---------------|--------------|-------------|
| Factual support     | 0.319          | 0.554         | 3.09e-64     | 0.576       |
| Hallucination score | 0.500          | 0.280         | 5.33e-51     | 0.492       |

95% CI for factual gain: [0.211, 0.258]

Table 2. H2: Code-mixing index (CMI) tertiles — grounded outcomes

| CMI level | $n$ | Mean CMI | Grounded factual | Factual gain |
|-----------|-----|----------|------------------|--------------|
| Low       | 385 | 0.351    | 0.554            | +0.202       |
| Medium    | 384 | 0.428    | 0.544            | +0.241       |
| High      | 396 | 0.493    | 0.563            | +0.260       |

Kruskal-Wallis on factual gain:  $H = 3.879$ ,  $p = 0.144$  (n.s.) | Spearman rho (CMI vs gain) = 0.070,  $p = 0.016$



**06**

# **Research Design.**

---

## *the research design...*

The study follows an **evidence-first RAG pipeline** for grounded Hinglish clinical **explanations**: query encoding, evidence retrieval, grounded generation, and (in later extensions) **optional multimodal context**. Code-switched queries are encoded with cross-lingual text representations so **mixed-language intent is preserved**.

The RAG core **retrieves ranked English clinical case** narratives from a curated multi-specialty corpus, then injects that text into generation to **limit reliance on parametric knowledge**. The generator returns Hinglish explanations conditioned on retrieved evidence, **supporting evidence-aligned outputs**.

Query–evidence alignment uses **real Hinglish clinical queries paired to English cases** via semantic retrieval (e.g. LaBSE + dense search) and condition-aware filtering, **rather than relying on synthetic query generation** as the main data path.

Multimodal evidence (e.g. imaging) **is reserved for extension of the same pipeline**, while the current design centers on text retrieval and text-conditioned generation.

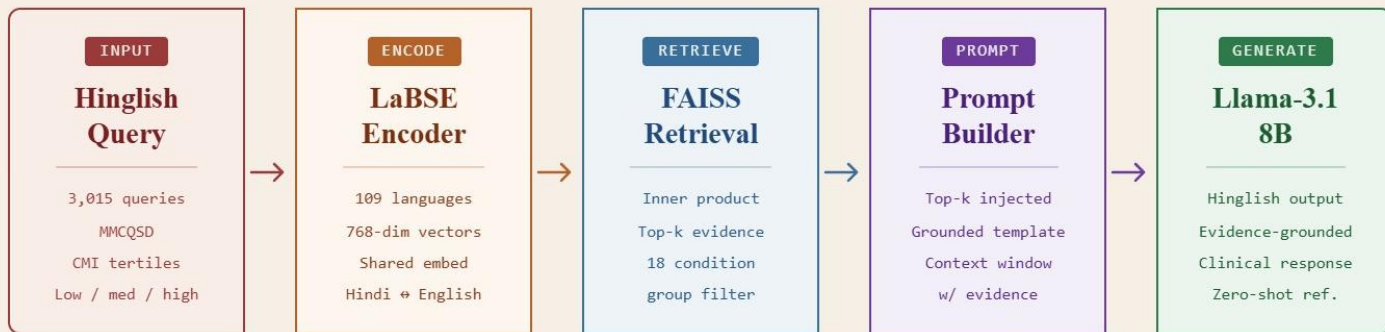


# the research design...



| ELEMENT            | DESCRIPTION / OUR APPROACH                                                                                                                                     |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type Of Research   | <b>Applied and Analytical:</b> evaluation of evidence-grounded Hinglish clinical decision support through a RAG-style pipeline.                                |
| Research Approach  | <b>Quantitative and Empirical:</b> paired comparison of zero-shot and text-grounded RAG outputs on the same Hinglish queries (H1, H2).                         |
| Design Strategy    | <b>Comparative Design</b> (zero-shot vs text-grounded RAG); text retrieval and generation are central, with multimodal inputs envisioned as a later extension. |
| Unit Of Analysis   | Each Hinglish query and its generated explanation, <b>judged against retrieved English case evidence</b> and, where relevant, code-mixing level.               |
| Data Sources       | <b>MMCQSD</b> (Hinglish queries) and <b>MultiCaRe</b> (English clinical case narratives); other public sets used only for background or early exploration.     |
| Validation Method  | <b>Paired scoring</b> across grounding modes and CMI groups, with non-parametric tests and effect sizes; H3 (evidence-type) reserved for follow-on work.       |
| Tools & Frameworks | Python; LaBSE and FAISS for retrieval; Groq (Llama-3.1-8B) for generation; Streamlit interface; Hugging Face ecosystem as supporting tooling.                  |

# Pipeline Architecture



## MultiCaRe Knowledge Base

LaBSE-INDEXED • FAISS INNER PRODUCT • ICD CODE FILTERED

**61,316**

clinical cases

**18**

condition groups

**PMC**

case reports

**ICD**

code filter

## Zero-shot Baseline

SAME LLM • NO EVIDENCE

Factual support **0.00**

Hallucination **1.00**

Evidence **none**

## EVALUATION FRAMEWORK

### CMI Stratification

Hindi token ratio  
Tertile split  
Low / med / high  
Stable across all

### Concept Scoring

24 clinical cats.  
Negation detect  
Output n evidence  
Token-level match

### Metrics Defined

Factual = output  
n evidence set  
Hallucin. = output  
\\ evidence set

### Results $p < 10^{-50}$

**+73.5%**

factual support

**-44.0%**

hallucination

# proposed prototype demonstration (streamlit)...

About

Architecture

1. Encode query with LaBSE (cross-lingual)

2. Retrieve top-k evidence via FAISS

3. Generate grounded Hinglish response

Components

- Encoder: sentence-transformers/LaBSE
- Index: FAISS (inner-product)
- Generator: Llama-3.1-8B-Instant (Grok)
- Evidence: MultiCaRe clinical cases

Retrieved evidence (top-k)

3

☒ Compare with zero-shot

Hinglish Medical RAG

localhost:8501

Deploy

## Grounded RAG for Hinglish Clinical Queries

Type a Hinglish patient query and the system will retrieve relevant clinical evidence from MultiCaRe and generate a grounded response using Llama-3.1-8B.

Try an example query

Meri beti ko ek rash hai jo theek nahi ho raha hai. Hydrocor...

Doctor meri eye me sujan hai aur laal ho gayi hai, bahut dar...

Mere gale me tonsils me dard hai aur khaana nahi kha pa rahi

Mujhe pair me sujan hai aur chalne me bahut takleef hoti hai

Meri skin pe ek growth aa gayi hai jo badhti ja rahi hai

Mere muh me ulcers hain aur kuch bhi khaane me jalan hoti ha...

Or type your own

Hinglish query

Doctor, meri skin pe rash hai aur bahut khujli ho rahi hai ...

Get Response

## Retrieved Evidence

> Evidence 1: eye\_redness | Similarity: 0.259 | PMC11014721

> Evidence 2: eye\_inflammation | Similarity: 0.254 | PMC5700511

> Evidence 3: eye\_redness | Similarity: 0.254 | PMC8784481

## Grounded Response

Bhai, tumhare case mein kuchh cheezon ka dhyan dena hoga. Pehle, tumhare age 72 hain aur tumhein vision loss ki samasya hai. Fundus examination mein tumhare right eye mein MH (macular hole) nikla hai. Iske alawa, tumhare intraocular pressure normal hai, lekin tumhare right eye mein visual acuity kam hai.

Humne tumhare case mein PPV (pars plana vitrectomy), ILM translocation, aur sulfur hexafluoride (SF6) gas injection kiya hai. Ab tumhari treatment ke baad tumhare symptoms kam honge.

Factual Support

0.25

Hallucination

0.00

## Zero-Shot Response

Arre bhai, aapka sujan aur laal hone ka sawal bahut chinta dene wala hai. Main aapko kuchh samajhne ka prayaas karta hoon. Aapke symptoms ka sambandh eye infection, conjunctivitis ya even korneal ulcers se ho sakta hai. Hum zaroori cheezon ka check-up karenge aur phir hi aapko sahi treatment di jaayegi.

Factual Support

0.00

Hallucination

1.00



**07**

**Data – Primary / Secondary.**

---



## *datasets utilized...*

### A. MMCQSD (Multimodal Medical Code-mixed Question Summarization)

**Description:** This dataset contains **3,015 samples** consisting of medical images (Skin, ENT, Eye, etc.) paired with patient queries written in Hinglish (Hindi-English code-mixed) and their **corresponding English summaries**.

**Relevance:** This is the **heart of our linguistic innovation**. It is the only open dataset that captures the "messy" way Indian patients describe symptoms. We can use it to **teach our model how to interpret** informal Hinglish and map it to clinical terms.

### B. MultiCaRe (Multilingual Case Repository–Style Clinical Case Corpus)

**Description:** Large open collection of **English clinical case narratives** from PubMed Central style open-access literature. The project uses a **specialty-filtered subset** aligned to MMCQSD condition groups so retrieved evidence matches the query domain.

**Relevance:** English **evidence base** for the RAG system: indexed case narratives ground Hinglish generation in retrieved clinical text, **reducing reliance** on the model's internal knowledge.

# data – primary & secondary...

## 1. Primary Data (Cross-lingual Grounding Resource)

**Dataset:** MMCQSD–MultiCaRe Query–Evidence PairedAlignment

**Description:** A **paired resource of 3,015 Hinglish clinical queries** from MMCQSD linked to English clinical case narratives drawn from a MultiCaRe-derived, specialty-filtered corpus. Pairing uses cross-lingual dense embeddings (LaBSE), FAISS similarity search, and condition-aware constraints so queries retrieve clinically compatible evidence.

**Rationale:** Constitutes the **main empirical basis for retrieval-augmented Hinglish generation** and **for paired comparison** of grounded versus zero-shot outputs on the same queries, including code-mixing–stratified analysis.

## 2. Secondary Data (Foundational Open Corpora)

**Source 1:** MMCQSD (MedSumm distribution)

**Rationale:** Authoritative **public release of real Hinglish medical queries** (and associated metadata) that anchors the project in authentic code-mixed patient language rather than synthetic query fabrication.

**Source 2:** MultiCaRe

**Rationale:** Large-scale, open **English clinical case-report text** for building a retrieval index; after specialty filtering & encoding, it supplies the evidence passages that constrain generation & support factuality-oriented evaluation.



**08**

# **Data Science Techniques.**

---

# *data science techniques implemented...*

## *1. Cross-Lingual Text Representation*

**Technique:** LaBSE (language-agnostic sentence embeddings).

**Application:** Encode Hinglish queries and English MultiCaRe case text in a shared 768-dimensional space so informal code-mixed queries retrieve semantically related clinical narratives.

## *2. Domain-Aware Dense Retrieval*

**Technique:** FAISS inner-product search with adaptive context truncation.

**Application:** Retrieve top-k passages, then trim the tail where similarity scores fall sharply (MMed-RAG-style), reducing irrelevant case text that could bias or destabilize generation.

## *3. Evidence-Conditioned Generation*

**Technique:** Retrieval-augmented prompting.

**Application:** Inject ranked excerpts into the prompt; Llama-3.1-8B-Instant via Groq produces Hinglish responses conditioned on retrieved evidence rather than unconstrained parametric completion.

## *4. Quantitative Grounding Evaluation*

**Technique:** Concept-level overlap metrics with negation handling; code-mixing index (CMI) stratification; non-parametric paired and group tests.

**Application:** Score factual support and hallucination relative to retrieved text and analyze robustness across low / medium / high code-mixing (H1, H2).

# *data science techniques planned...*

## 5. *Preference-Based Alignment*

**Technique:** Direct preference optimization (DPO) or equivalent preference learning.

**Application:** Train or align the generator to favor evidence-consistent Hinglish answers over unsupported alternatives, once a suitable preference data and compute pipeline are in place.

## 6. *Parameter-Efficient Medical Adaptation*

**Technique:** QLoRA (or similar PEFT) on a medical or dialogue backbone.

**Application:** Adapt model behavior to clinical Hinglish style and safety norms under GPU constraints, extending the current frozen API generator setup.

## 7. *Multimodal Evidence And Metrics*

**Technique:** BioMedCLIP-class image–text encoders; multimodal RAG; MMFCM or related factuality measures.

**Application:** Retrieve and condition on imaging alongside text, and evaluate multimodal grounding when the corpus and hardware support full multimodal indexing and inference.

These items **outline a forward research agenda**, with preference learning, multimodal evidence, stronger evaluation, and evidence-authority comparisons, which are **contingent on resource availability and formal study design** beyond the present text-grounded system.

**09**

# **Data Analysis.**

---

# the data analysis...



Our analyses examine **how alternative system configurations perform** under linguistic and evidence-related manipulations, not only under aggregate accuracy-style summaries.

The objective is to **characterize how grounding choices affect reliability, robustness, and stability** of Hinglish clinical explanations as code-mixing intensity and context format vary.

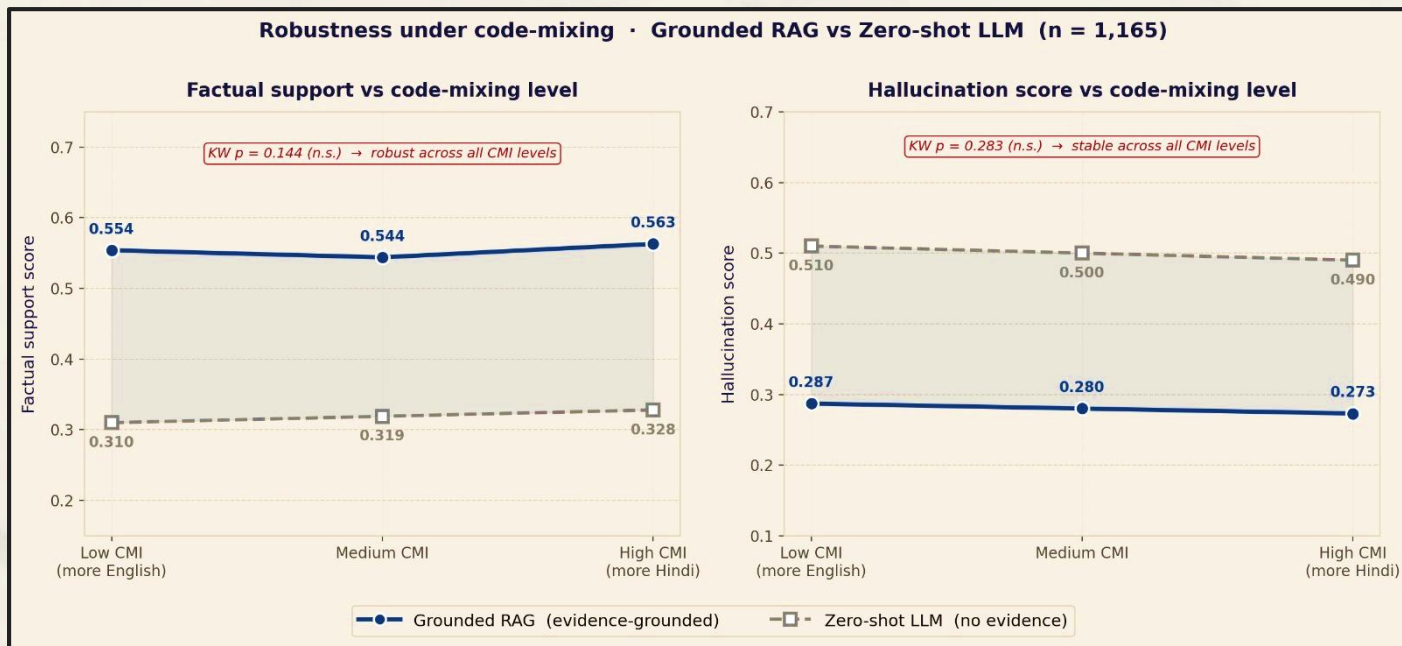
| ANALYSIS DIMENSION | COMPARISON                                          | PURPOSE                                                      |
|--------------------|-----------------------------------------------------|--------------------------------------------------------------|
| Generation         | Zero-Shot vs. Grounded RAG                          | Does grounding help facts and cut hallucinations? (H1)       |
| Code-Mixing        | Low / Mid / High CMI                                | Does grounding still work when Hinglish is heavier? (H2)     |
| Evidence Format    | Raw Case Text vs. Structured Extract                | Does structuring evidence help before generation? (ablation) |
| Retrieval Context  | Top-k + Adaptive Cut                                | Keep relevant passages, drop weak tail matches.              |
| Future (Planned)   | Clinical vs. General Text;<br>Text vs. Text + image | H3 + multimodal when data and setup are ready.               |



# the data analysis...



Queries are **stratified by code-mixing intensity** (CMI tertiles: low, medium, high). Paired zero-shot and text-grounded RAG outputs are **contrasted within each stratum**; cross-stratum differences are assessed with Kruskal–Wallis, Mann–Whitney U and Spearman correlation, **evaluating stability of grounded performance** under increasing Hindi–English mixing.



**10**

## **Limitations and Future Scope.**

---

# the limitations...



- 1) **Linguistic Scope:** results pertain to Hindi–English code-switching only; extension to other Indian code-mixed languages or fine-grained dialect variation is untested.
- 2) **Modality & Evidence Scope:** the system is text-grounded on open-access English clinical case narratives (MultiCaRe-derived corpus), not on live EHR data, structured labs, or imaging workflows; multimodal evidence lies outside the reported evaluation.
- 3) **Retrieval Sensitivity:** minor rephrasing or mixing patterns can change dense retrieval rankings, inducing run-to-run variability in injected evidence and in generated text.
- 4) **Model Adaptation & Operations:** the LLM is not clinically fine-tuned or preference-optimized; generation relies on a hosted API and network connectivity, with trade-offs for latency, cost, and deployment in resource-constrained settings.
- 5) **Evaluation Validity:** outcomes rest on automated, concept-level alignment scores and hallucination proxies, not on clinician judgment of safety, usefulness, or trust; assessment is offline and does not incorporate individual patient history, comorbidities, or longitudinal care context.
- 6) **Decision-Support Positioning:** outputs are intended as interpretive assistance only and do not constitute diagnosis or treatment recommendation; step-by-step clinical reasoning is not explicitly surfaced for audit.

# the future scope...



- 1) **Extension to Other Code-mixed Languages:** the text-grounded RAG template can be ported to additional Indian code-mixed registers (e.g. Tamil–English, Bengali–English) through new query corpora, encoder or lexicon adaptation, and re-calibrated evaluation.
- 2) **Multimodal & Longitudinal Clinical Context:** image–text retrieval (e.g. BioMedCLIP-class encoders) and integration of laboratory values, vitals, waveforms, or longitudinal EHR snippets would extend the current case-narrative index toward richer grounding under complex presentations.
- 3) **Parameter-Efficient Adaptation & Preference Learning:** QLoRA (or similar PEFT) on medical dialogue backbones, together with DPO or related preference optimization, could specialize Hinglish style and reduce unsupported content beyond prompt-only API generation.
- 4) **Human-in-the-Loop Clinical Validation:** prospective studies with clinicians and lay users are needed to assess readability, trust, perceived safety, and workflow fit beyond offline automated scores.
- 5) **Personalization, Governance & Auditability:** future systems could condition explanations on health literacy, language preference, and region-specific usage, while embedding regulatory constraints, risk-aware disclosure, and retrieval–generation audit trails for accountable deployment.
- 6) **Evidence-Provenance Comparisons (H3):** systematic paired contrasts between clinical case text and general biomedical retrieval sources would test whether evidence authority changes grounded Hinglish quality.

**11**

**Conclusion.**

---

## *the conclusion...*



This **study supports the view** that clinical reasoning under code-switched Hinglish inputs is vitally a **grounding & evidence-alliance problem**, not a translation problem.

Evidence-first, text-grounded Retrieval-Augmented Generation aligns informal Hinglish intent with retrieved clinical text, yielding **higher evidence-aligned factual support** and **lower hallucination-oriented scores** than zero-shot baselines on paired queries (H1).

Empirical results indicate **stable grounded performance across code-mixing intensity** (CMI tertiles) without statistically detectable loss of grounding benefit across those strata (H2); the evaluated system remains text-only, **positioned as decision support rather than autonomous diagnosis**.

Overall, the work underscores that **evidence-grounded reasoning is central** to safer, more reliable, and more inclusive clinical AI assistants **in Indian code-mixed healthcare communication**, within the scope of offline, metric-based evaluation.

# the end, thank you! :)

---

updated research proposal of group 5 :

devika jonjale, B045  
saachi shinde, B048  
manjiri apshinge, B061

